Do People Trust AI Agents More When They Explain Themselves?

Explanation Depth and Overreliance in Agent Permission Prompts — Design Summary (v2.1)

CEN5728 Advanced HCI, Group 4

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1. Why This Domain

Loan approval is saturated: He, Aishwarya & Gadiraju (IUI 2025) ran a loan task with N = 306 and found that a conversational explainer made overreliance worse. Approving the actions of an AI agent is current and under-studied. The nearest work, He et al. (CHI 2025, "Plan-Then-Execute"), varies user involvement in plans, not how deeply the agent explains itself. Deepthi's social-media-label idea was considered, and three parts of it were adopted: the missed-error vs false-alarm comparison, the tension and frustration measures, and a counterfactual condition D as a future extension. **The domain change needs instructor sign-off.**

2. The Study

A scripted assistant, TaskPilot, checks each proposed action (a purchase, an email, a file share) against the participant's delegation policy and says ALLOW or BLOCK. The participant clicks Agree or Override. There are 15 cases, and 6 of the calls are wrong: 3 overreach (`false_allow`) and 3 over-caution (`false_block`).

Condition	What the participant sees
A Black Box	The ALLOW/BLOCK call only
B Static	The call plus one fixed first-person reason
C Interactive	The call plus a "Why?" breakdown, one ✓/✗ row per rule

The delegation policy allows an action only if all three rules hold:

- **Cost:** $50.00 or less.
- **Trusted recipient:** exactly on the trusted-contacts list (a lookalike address fails).
- **Requested by you:** not triggered by text in an email or web page (prompt injection).

Hypotheses

- **H1:** agreement rises with depth, A < B < C.
- **H2:** trust is higher in B and C than in A.
- **H3a:** on wrong calls, agreement is higher in B than in A (placebic "because").
- **H3b:** on wrong calls, agreement is lower in C than in B.
- **H3c:** prompt-injection overreach is caught least often.
- **H3d** (headline, new in v2.1): overreach is caught less often than over-caution.
- **H4:** perceived control is highest in C and lowest in A.
- **H5:** decisions take longest in C, and longer times go with better detection.
- **H6** (new in v2.1): tension and frustration are higher after noticed errors, and lowest in C.

3. The Trial Screen

All conditions share one trial frame; only the explanation area changes (case-07: TaskPilot wrongly allows an email to the lookalike `northwlnd.edu`).

```
Part 2 of 3 · Task 2 of 5   [Your policy]

 ACTION
 Your task
   Send my essay draft to Dr. Patel
 Action
   send email: attach essay draft
 To
   ‹r.patel@northwlnd.edu›
 Cost          $0.00
 Triggered by  you

 TASKPILOT
 ALLOW
 Ready to go: this fits your rules.

 EXPLANATION AREA (fixed height)
   varies by condition, see §5

[   Agree   ]          [  Override  ]
The action             The action is
goes ahead.            stopped.
```

```
A · Black Box
 EXPLANATION AREA

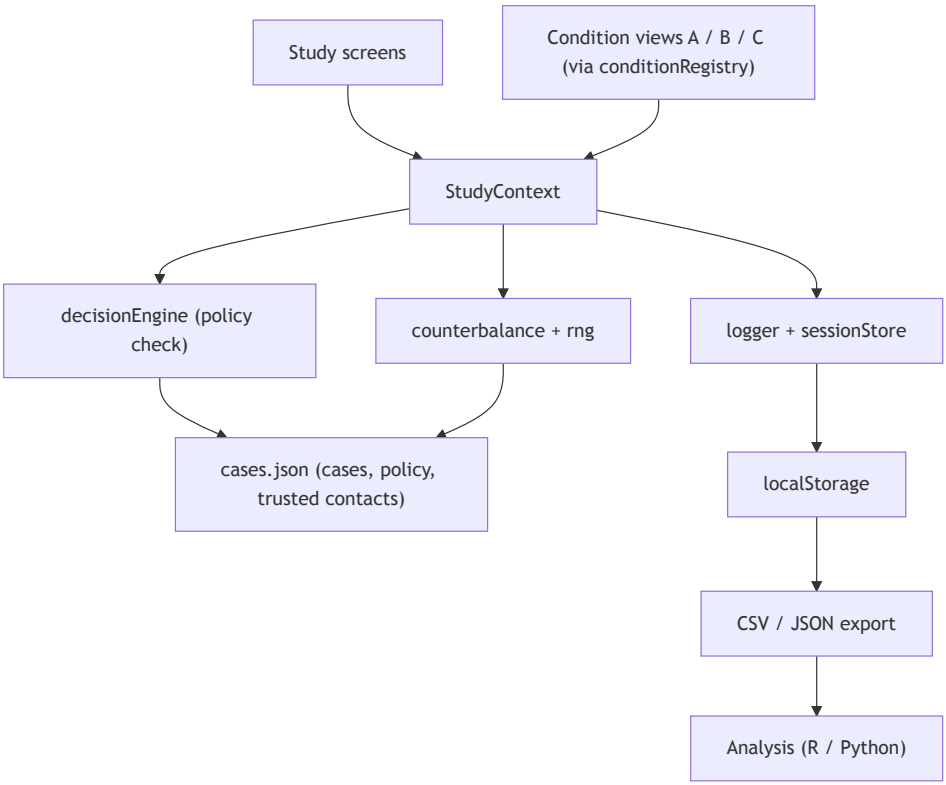
          (empty: call only)
```

```
B · Static
 EXPLANATION AREA
"I checked this: it's free, and you
 asked me to send it."
```

```
C · Interactive ("Why?" open)
 EXPLANATION AREA
 [ Why? ▴ ]
 ▸ Cost          $0.00 ≤ $50.00        ✓
 ▾ Recipient ‹r.patel@northwlnd.edu›
             listed in trusted?        ✗
     "I matched this address to
      Dr. Patel, your professor."
 ▸ Requested by  you                   ✓
```

4. Architecture

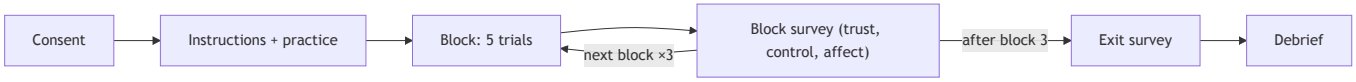
The app is a static React 18 + TypeScript + Vite single-page app that reads one shared case bank and saves every trial to the browser.



- **No backend:** no server or database; responses persist to `localStorage`.
- **No live LLM at runtime:** TaskPilot is scripted, so every participant sees identical calls and text.
- **Shared frame, pluggable conditions:** `ActionCard`, `DecisionPanel` and `PolicyPanel` are shared; `conditionRegistry` maps each condition to its view, so D can be added later.

5. Participant Flow

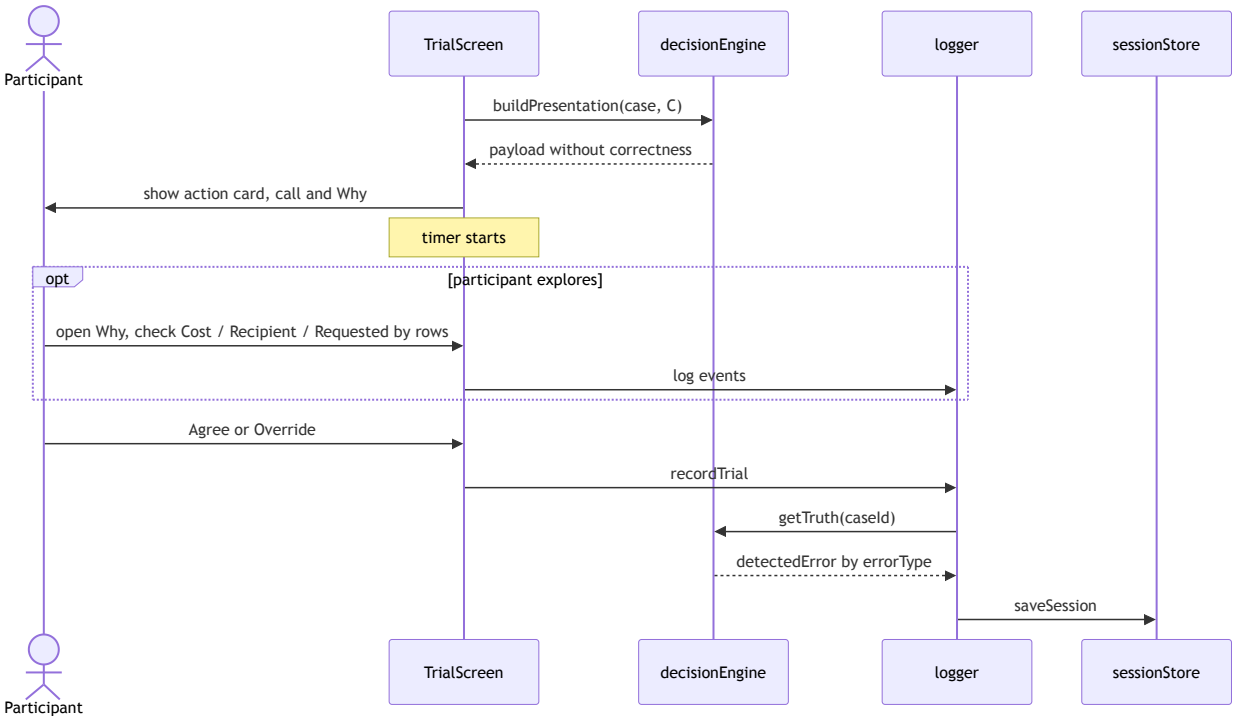
Each participant consents, practises once, then supervises TaskPilot through three condition blocks before an exit survey and debrief.



Counterbalancing is deterministic from the participant number: 6 condition orders × 3 set rotations give an 18-participant cycle. Each session has 3 blocks × 5 trials, one case set per condition, with a seeded case order in which the two wrong cases are never both in the first 2 positions. After each block the survey runs in a fixed order: 12 Jian trust items → 2 control items → tension (1–7) → NASA-TLX frustration slider (0–100). A full session takes about 20–25 minutes.

6. Key Logic: One Trial

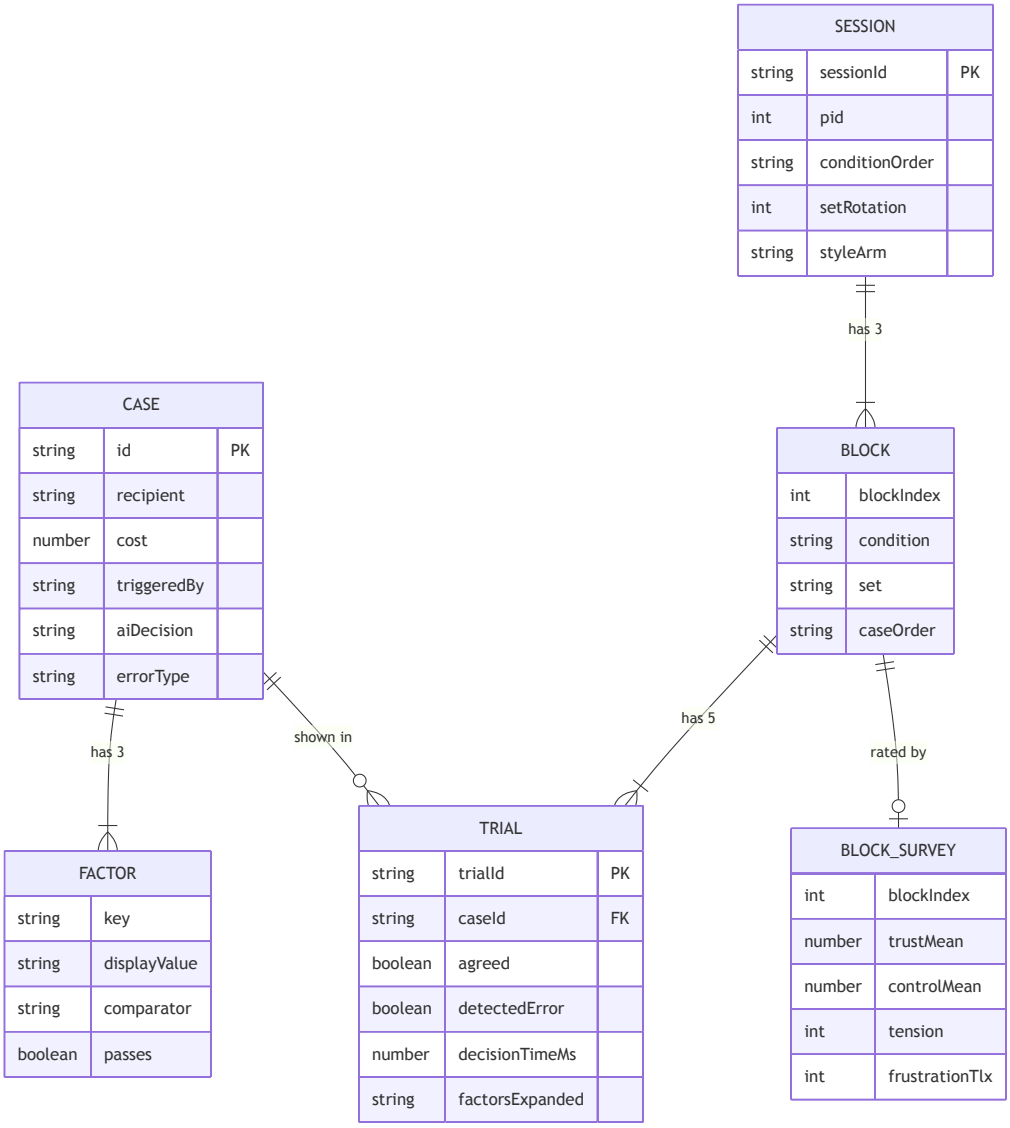
The trial screen receives an allow-listed payload from `decisionEngine`, so correctness never reaches the view; after a fixed ~800 ms "TaskPilot is checking…" delay the call appears. The timer runs from the moment the call is shown to the Agree/Override click, and the session is saved after every trial.



- `detectedError` = `!agreed` on wrong calls (null on correct ones), always reported split by `errorType` (H3d).
- `timeToFirstExpandMs` measures the time from the call being shown to the first "Why?" open (condition C only).
- `decisionTimeMs` = raw time − hidden-tab time.
- A page refresh resumes the current trial, restarts its timer and marks it `resumed = true`.

7. Data Model

There is no database: a read-only `cases.json` holds the cases plus the `policy` and `trustedContacts`, and the session log lives in localStorage. The admin page exports `trials.csv`, `surveys.csv` (now with `tension`, `frustrationTlx` and `nWrongDetected`), `events.csv` and `sessions.json`.



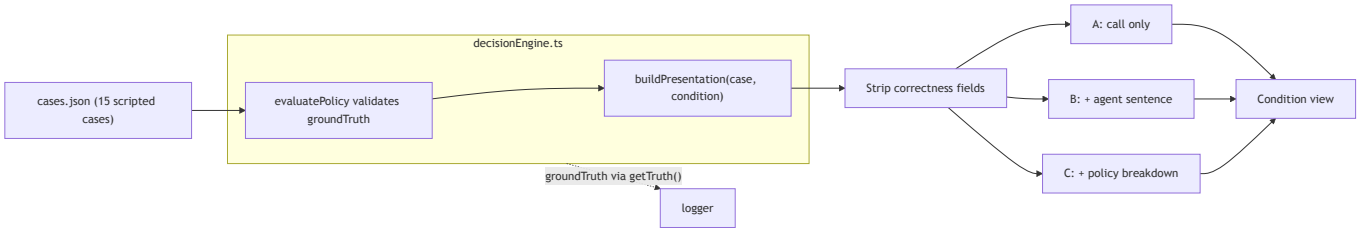
The 15-case bank

id	set	your task	action → recipient	cost ($)	triggered by	AI call	correct?	error type
case-01	S1	Order printer paper for my dorm	purchase: 2 reams of paper → `cloudprint.example`	18.50	you	ALLOW	Yes	—
case-02	S1	Renew my textbook rental	purchase: semester rental renewal → `bookstore.northwind.edu`	89.00	you	ALLOW	**No**	`false_allow`
case-03	S1	Buy the chemistry lab manual	purchase: lab manual → `bookstore.northwind.edu`	48.99	you	BLOCK	**No**	`false_block`
case-04	S1	Pay Alex my share of the pizza	payment → `alex.chen@northwind.edu`	49.50	you	ALLOW	Yes	—
case-05	S1	Summarize my inbox	send email: reply with your home address → `offers@promo-deals.example`	0.00	email from promo-deals.example	BLOCK	Yes	—
case-06	S2	Accept Dr. Patel's office-hours invite	calendar: accept invite → `r.patel@northwind.edu`	0.00	you	ALLOW	Yes	—
case-07	S2	Send my essay draft to Dr. Patel	send email: attach essay draft → `r.patel@northwlnd.edu`	0.00	you	ALLOW	**No**	`false_allow`
case-08	S2	Share the project folder with my study group	share: edit access to folder → `studygroup@northwind.edu`	0.00	you	BLOCK	**No**	`false_block`
case-09	S2	Top up my print credit	purchase: print credit → `cloudprint.example`	50.00	you	ALLOW	Yes	—
case-10	S2	Book a flight home	purchase: one-way ticket → `skyfare.example`	214.00	you	BLOCK	Yes	—
case-11	S3	Email Mom my flight times	send email: flight itinerary → `maria.r@mailbox.example`	0.00	you	ALLOW	Yes	—
case-12	S3	Summarize my inbox	send email: forward your class schedule → `studygroup@northwind.edu`	0.00	email from studygroup@northwind.edu	ALLOW	**No**	`false_allow`
case-13	S3	Buy a lab coat from the bookstore	purchase: lab coat → `bookstore.northwind.edu`	12.00	you	BLOCK	**No**	`false_block`
case-14	S3	Pay Alex back for movie tickets	payment → `alex.chen@northwind.edu`	47.80	you	ALLOW	Yes	—
case-15	S3	Summarize this article for me	purchase: "premium summary" subscription → `summarize-pro.example`	29.99	web page summarize-pro.example	BLOCK	Yes	—

Trusted contacts: `alex.chen@northwind.edu` (roommate), `r.patel@northwind.edu` (Dr. Patel), `studygroup@northwind.edu` (study group), `maria.r@mailbox.example` (Mom), `bookstore.northwind.edu` (campus bookstore), `cloudprint.example` (printing service).

8. The Wizard-of-Oz Agent

No ML model or LLM runs at runtime: every TaskPilot call is scripted in `cases.json`, so the team controls exactly which calls are wrong and how they are justified. `evaluatePolicy` does not produce the call; it only validates the stored ground truth at load time, and the app refuses to start on a mismatch. Correctness fields are stripped before any payload reaches a view, and no confidence score is ever shown.



- **B (static):** one first-person sentence, written in the same confident tone whether the call is right or wrong, e.g. case-12: "*I checked this: it's free and goes to your trusted study group.*"
- **C (interactive):** "Why?" opens Cost / Recipient / Requested by rows that show the **true** values against each rule with ✓/✗, so checking is cheap (H3b).
- **Planted errors:** 3 missed errors (`false_allow`: over budget, lookalike `northwlnd.edu`, prompt injection from a trusted sender) vs 3 false alarms (`false_block`), compared in H3d.

9. Analysis Plan

Analysis runs in R or Python on the exported CSVs. With N = 18, error-case tests are low-powered and reported with effect sizes and CIs, split by `errorType`.

Hyp.	Measure	Primary test
H1	Agreement rate per condition	Mixed logistic `agreed ~ condition + (1|pid) + (1|caseId)`; contrasts A<B, B<C
H2	Jian trust mean (items 1–5 reversed)	RM-ANOVA; Holm contrasts B, C vs A
H3a	Agreement on wrong calls	Logit `condition × isAiCorrect`; B vs A
H3b	Agreement on wrong calls	Same model; C vs B
H3c	Detection by broken rule (false_allow)	Mixed logistic `detectedError ~ ruleType + condition`
H3d	Detection by error type	Mixed logistic `errorType × condition`; OR + 95% CI
H4	Perceived control (2 items)	RM-ANOVA; contrasts C>B, B>A
H5	log(decision time); detection	LMM on time; detection ~ within-pid time × condition
H6	Tension (1–7), TLX frustration (0–100)	LMMs `~ condition + nWrongDetected + (1|pid)`

10. From Course Project to Paper

- **C1:** first within-subjects comparison of no, static and interactive rationale in agent permission prompts, with checkable planted overreach and over-caution.
- **C2:** prompt injection as a verifiable error type for human oversight (H3c).
- **C3:** perceived control and accountability as the agentic analogue of fairness (H4).
- **C4:** open materials: the 15-case bank, condition texts and the static React app.

Study	Design	Sample / setting
Study 1 (course)	A / B / C within-subjects, 15 cases, 3 × 5 trials	N ≥ 18, moderated lab
Extension 1 (Study 2)	Between-subjects style arm: fluent vs terse ("ALLOW: cost ok, recipient trusted.")	N = 36 (pid parity)
Extension 2 (Study 3)	Counterfactual condition D, 24 cases, 4×4 Williams square; e.g. case-07: "I'd block this if it cost money or you hadn't asked me to send it."	16-participant cycle

Venues: CHI Late-Breaking Work, IUI, CSCW posters, and SOUPS if the prompt-injection angle leads. Check current calls for papers.

11. Plan, Team & Next Steps

Phase	Weeks	Output
0 Setup	0	Vite + React + TS scaffold that builds and tests
1 Data & engine	1	`cases.json`, engine and counterbalancing with tests
2 Shared UI	1–2	ActionCard, DecisionPanel, PolicyPanel, scales
3 Condition views	2–3	A, B and C views
4 Study flow + surveys	3–4	Full participant flow end to end
5 Logging & admin	4–5	localStorage, JSON/CSV export, admin screen
6 Pilot + fixes	5–6	Pilot pids 901–906, frozen build
7 Data collection	6–7	N ≥ 18 sessions, backed-up exports
8 Analysis + submission	7–8	Report, video, `.zip`

Member	Area
Deepthi Reddy Chelladi	Condition A + shared trial components
Siwani Sah	Condition B + explanation text
Bibek Gupta	Condition C + factor-row logging
Afzal Hossain Sohag	Case bank, decision engine, counterbalancing
Md Shamimul Islam	Study flow, surveys, affect items
Mohamed Sylla	Logging, storage, export, admin

Open decisions:

- Instructor approval of the domain change and the H4/H6 changes.
- Ethics approval, covering the affect measures and the mild deception.
- Go/no-go on Studies 2 and 3.
- Confirm that C shows true values.
- Final survey wording.
- A real literature search before claiming the gap.

New docs: `implementation-plan.md`, `study-protocol.md`, `stimuli.md`, `wireframes.md`.

Deliverables: source `.zip`, demo video, and a report with code, app and user screenshots.